CHAPTER – 2

RELATED WORKS

2.1 LITERATURE SURVEY

The studies on hate speech and offensive content detection employ various methodologies and datasets to address different aspects of the problem. In [1], Roy et al. (2024) proposed a Long Short-Term Memory (LSTM)-based approach to detect hateful sentiment in real-time tweets, using the Hatebase Twitter dataset. The authors highlight LSTM's efficacy in capturing sequential patterns in text data, essential for hate speech detection. They also experimented with a comparative feature set approach, enhancing the model's robustness. However, the model's high computational cost poses challenges for scaling to larger, real-time datasets. Ahmed and Lin (2022) in [2] introduced deep explainable active learning for hate speech detection on various social media datasets, including Twitter. Their approach focuses on explainability, improving model transparency, which is crucial for ethical AI applications in sensitive areas like hate speech detection. However, the complexity of active learning hinders real-time application, as the model requires iterative updates with new labeled data. In contrast, Alsafari et al. (2020) in [3] developed a model tailored for Arabic social media, addressing unique linguistic nuances with specific NLP preprocessing techniques for Arabic text. This model, while effective in detecting hate speech within Arabic social media, faces limitations in cross-linguistic applicability due to its language-specific dataset and preprocessing methods.

The following studies offer a range of methodologies and datasets for hate speech detection, addressing challenges across languages, feature representations, and model structures. Al-Hassan and Al-Dossari (2019) in [4] conducted a survey on multilingual hate speech detection, focusing on corpora across various languages to identify best practices for generalizing hate speech models. This survey emphasizes the importance of multilingual datasets for addressing linguistic diversity, although it also highlights the scarcity of high-quality, labeled data across non-English languages. In [5], Badjatiya et al. (2017) explored deep learning approaches for hate speech detection on Twitter, using embeddings with a range of deep learning models including LSTM and Convolutional Neural Networks (CNN). This approach benefits from the powerful representation capabilities of embeddings but requires high computational resources, limiting real-time application potential. Bölücü and Canbay (2021) in [6] introduced Graph Convolutional Networks (GCNs) for hate speech and offensive content detection, which utilize relational information between words and users to improve context-aware feature learning.

While GCNs excel in capturing intricate relational data, their performance can be sensitive to graph structure quality and large-scale deployment challenges.

These studies employ a variety of methodologies to tackle abusive and hateful content detection, using distinct approaches and datasets that highlight specific challenges and advances in the field. Nobata et al. (2016) in [7] explored abusive language detection in online user-generated content, applying linguistic and syntactic features along with machine learning models like logistic regression and SVM. The study uses Yahoo's online comments dataset, showing effectiveness in identifying abusive content but facing challenges in generalizing to other platforms due to dataset specificity. Malmasi and Zampieri (2017) in [8] examined hate speech detection in social media through traditional machine learning, using an ensemble approach with multiple classifiers on social media datasets. This approach provides reliable performance, but it relies heavily on high-quality labeled data, which can be limited across languages and topics. Aluru et al. (2020) in [9] focused on multilingual hate speech detection using deep learning models, including BERT and multilingual embeddings, to accommodate varied languages within the same model. This approach is advantageous for cross-linguistic applications, yet it remains resource-intensive and less interpretable, posing challenges in practical deployments for real-time monitoring.

Mandl et al. (2021) in [10] provided an overview of the HASOC subtrack at FIRE 2021, which addresses hate speech detection in English and Indo-Aryan languages. The study used multilingual datasets and benchmarked several approaches to identify language-specific challenges and evaluate model performance in multilingual settings. This work highlights the need for language-specific preprocessing but reveals limitations in achieving generalization across languages. In [11], Baziotis et al. (2017) applied deep LSTM with attention for sentiment analysis on SemEval-2017's dataset. While focused on sentiment rather than hate speech, the attention mechanism allows for improved context capture, showing potential applicability in hate speech detection. However, the approach is computationally intensive, and LSTM models may struggle with large datasets. Nayla et al. (2023) in [12] utilized BERT for hate speech detection on Twitter, which leverages pretrained embeddings to handle nuanced language variations. BERT's language understanding capabilities allow for strong results, especially in capturing context, but the model requires substantial computational resources and may face challenges with real-time detection on large datasets.

King and Sutton (2013) in [13] analyzed the temporal clustering of hate crimes, focusing on offline hate-motivated offenses through criminological modeling. Their research highlights patterns in hate crime occurrences over time, suggesting that certain social or political events may trigger spikes in hate crimes. However, this study's offline focus limits its direct applicability to online hate detection. Burnap and Williams (2015) in [14] applied machine classification and statistical modeling to identify cyber hate on Twitter, using social media data to inform policy and decision-making. This study demonstrates the potential of machine learning for practical applications in policy but is limited by the model's reliance on platform-specific language patterns, which can reduce its effectiveness on other social media platforms. In [15], Warner and Hirschberg (2012) focused on detecting hate speech across the web using a language model based on lexical and syntactic features. This approach captures explicit hate speech effectively but struggles with subtler, implicit language forms and is limited by the scope of its language model.

In [16], Kumar et al. (2023) proposed a system for emotion detection and emoji display using machine learning, aiming to map detected emotions to suitable emojis. Their model emphasizes ease of communication through visual cues but faces challenges in accurately capturing subtle emotional differences due to limitations in labeled emotional datasets. Praveen et al. in [17] focused on facial emotion detection with CNN for emoji creation, employing convolutional neural networks to identify facial expressions and translate them into emoji representations. While CNNs are effective in visual data processing, this approach may struggle with real-time applications due to high computational demands. In [18], Srivastava et al. (2021) utilized deep learning for emotion-based emoji retrieval, applying a model that retrieves emojis based on detected emotions, potentially enhancing communication efficiency. However, their approach is limited by the quality of emotion representation and the model's dependency on accurate facial cues, which can be impacted by variations in facial features and lighting.

In [19], Chrismanto et al. (2023) proposed a system for spam comment detection using emoji features and post-comment pairs, combined with ensemble machine learning methods. This approach improves spam detection accuracy by considering emojis as additional features that can convey sentiment, but it faces challenges in generalizing to different types of spam or platforms where emoji usage varies. Purver and Battersby (2012) in [20] experimented with emoticons in task-oriented dialogues, examining their impact on conversational context and how they can influence user intentions in dialogue systems. This study highlights the nuanced role of emoticons in enhancing human-computer interaction but suggests limitations in tasks requiring precise information or formal communication. Felbo et al. (2017) in [21] utilized millions of emoji occurrences to develop models for detecting sentiment, emotion, and sarcasm across any domain, using large-scale data to improve emotion detection. While effective in identifying emotional and sarcastic content, their approach depends on large datasets and may struggle with domain-specific or nuanced language, as emoji meanings can vary across contexts.

Suttles and Ide (2013) in [22] explored the emoticon-based detection of sarcasm on Twitter, using emoticons as key indicators for identifying sarcastic comments. Their approach highlights the utility of emoticons in sarcasm detection, though it struggles with the ambiguity and varied interpretations of emoticons in different contexts. Dalavi et al. (2024) in [23] proposed a method for hate speech detection using emoji-based classification combined with Bi-LSTM and GloVe embeddings. This method aims to enhance hate speech detection by incorporating emojis as features, capturing emotional nuances in text, and utilizing Bi-LSTM for better context understanding. While effective in recognizing nuanced hate speech, the method may face challenges in real-time detection and scalability due to the computational intensity of Bi-LSTM models. Felbo et al. (2017) in [24], using millions of emoji occurrences, developed models to detect sentiment, emotion, and sarcasm. Their study demonstrated how emoji-based data can improve detection in any domain by learning domain-independent representations of emotions and sentiment, although it may encounter challenges with specific domains or nuanced cultural interpretations of emojis.

Vayadande et al. (2023) in [25] proposed a mood detection and emoji classification system using tokenization and Convolutional Neural Networks (CNNs). Their model leverages CNNs to detect mood from textual input and predict corresponding emojis, highlighting the utility of deep learning for understanding emotional context in text. However, CNNs may face challenges with smaller datasets or overly generalized text, where mood nuances are less clear. Tare et al. (2023) in [26] explored hand-gesture-based emoji identification and chat integration, developing a system where hand gestures are interpreted to suggest emojis in chat applications. This system improves user interaction by using non-verbal cues, but practical deployment may be limited by the need for specialized hardware or high-quality gesture detection. Gupta et al. in [27] focused on hand gesture recognition for emoji prediction, applying gesture recognition technology to map gestures to emojis, enhancing user interaction in systems that require touchless control. While innovative, the approach may encounter challenges in real-time gesture tracking and in environments with varying lighting or background conditions.

Goh and Kulathuramaiyer (2020) in [28] developed an indigenous cultural values-based emoji messaging system, aiming to integrate culturally relevant symbols into digital communication. This approach emphasizes the importance of considering local cultural contexts when designing messaging systems, enabling more meaningful and contextually appropriate emoji usage. However, challenges include ensuring widespread adoption and balancing cultural specificity with global accessibility in diverse digital environments. Tandyonomanu (2018) in [29] examined emojis as representations of nonverbal symbols in communication technology, discussing how emojis function as a form of nonverbal communication that enhances text-based interaction by conveying emotions, actions, or ideas without words. This study underscores the significance of emojis in modern communication but also raises concerns about the potential for misinterpretation due to the lack of standardized emoji meanings across cultures.

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